

महाराष्ट्र वन सेवा मुख्य परीक्षा - 2024 दिनांक - ११ मे, 2024



2024

B20

BOOKLET NO.

240076

Forest Services
Zoology

Time Allowed : Three Hours

Maximum Marks : 200

Medium : English

Type of Paper : Conventional

Question Paper Specific Instructions

Please read each of the following instructions carefully before attempting questions :

1. There are **EIGHT** questions divided in two Sections, out of which **FIVE** are to be attempted.
2. Questions no. 1 and 5 are compulsory. Out of the remaining questions, **THREE** are to be attempted choosing at least **ONE** question from each Section.
3. The number of marks carried by a question/sub question is indicated against it.
4. Keep in mind the word limit indicated in the question if any.
5. Wherever option has been given, only the required number of responses in the serial order attempted shall be assessed. Unless struck off, attempt of a question shall be counted even if attempted partly. Excess responses shall not be assessed and shall be ignored.
6. Candidates are expected to answer all the sub-questions of a question together. If sub-question of a question is attempted elsewhere (after leaving a few page or after attempting another question) the later sub-question shall be overlooked.
7. Any page or portion of the page left blank in the Answer Booklet must be clearly struck off.
8. Unless otherwise mentioned, symbol and notation have their usual standard meanings. Assume suitable data, if necessary and indicate the same clearly.
9. Neat sketches may be drawn, wherever required.
10. The medium of answer should be mentioned on the answer book as claimed in the application and printed on admission card. The answers written in medium other than the authorized medium will not be assessed and no marks will be assigned to them.

Note : Candidates will be allowed to use Scientific (Non-programmable type) calculators.

P.T.O.

SEAL

**SECTION - A**

Q1. Write Short notes on **any five** from seven. **(8×5=40)**

- (a) Polymorphism in coelenterates.
- (b) Types of scales in pisces.
- (c) Ozone layer depletion.
- (d) Biological Rhythm.
- (e) Pearl culture.
- (f) Write a short note on Null-hypothesis.
- (g) Scanning Electron Microscope.

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- Q2.** (a) Explain in brief the comparative account of urinogenital system of male Labeo and male frog. **15**
- (b) Define population. Write in brief the characteristics of population. Add a note on population stability. **15**
- (c) What is the significant role of hormones and pheromones in animal behaviour ? **10**

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- Q3.** (a) Give general features of Aves. How the skeletal system helps in flight adaptation in birds ? Illustrate your answer with diagram. **15**
- (b) Define infectious disease. Describe the causative agent, symptoms, transmission diagnosis and control measures of COVID-19. **15**
- (c) What is central tendency ? What are the importance and measures of central tendency ? **10**

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- Q4.** (a) With a well labelled diagram, give an account of respiration in Arthropoda (Prawns, Cockroach and Scorpions). **15**
- (b) What is the impact of *Pyrilla Perpusilla* and *Sitophilus oryzae*, how it affect the production or yield ? Add a note on control measures of both. **15**
- (c) What is the Principle of Spectrophotometry ? What are the key challenges in analysis ? **10**
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**SECTION - B**

Q5. Write short notes on **any five** from seven. **(8×5=40)**

- (a) Describe structure and functions of mitochondria.
- (b) Define family pedigree. Draw various signs used in pedigree. Add a note on sickle cell anemia.
- (c) Describe Darwin's theory of natural selection.
- (d) Explain in brief Glycolysis.
- (e) Describe ultra structure of sperm and explain process of Spermatogenesis.
- (f) What is aging ? Explain in brief morphological, cellular and physiological changes in aging.
- (g) Describe in detail structure of lipid bilayer with membrane protein diffusion.

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- Q6.** (a) Describe double helical structure of DNA in detail. Add a note on replication of DNA. **15**
- (b) What is induced mutations ? Explain Muller experiment for radiation induced mutation. **15**
- (c) Explain enzyme classification with suitable example. **10**

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- Q7.** (a) Describe the ultrastructure of nephron and explain the mechanism of urine formation. **15**
- (b) What is placenta ? Describe the types of placentae based on degree of intimacy between the foetal and maternal tissue and explain the physiological role of placenta. **15**
- (c) Describe in detail Hardy-Weinberg equilibrium. Add a note on effect of violation of the Hardy-Weinberg assumptions. **10**

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- Q8.** (a) Describe in detail different types of chemical mutagens and their mechanism of action on DNA. **15**
- (b) What is respiration ? Explain physiology of gases exchange. **15**
- (c) What is test tube baby ? Describe various steps involved in it. Give its significance. **10**
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SEAL